PROJECT PROPOSAL (SCOPE DOCUMENT)

Web Based Interactive Quiz

Platform for Students

SUBMITTED BY
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SESSION
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Department of Information Technology
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Project Brief

This project is a **modern online quiz portal** designed specifically for university students to assess their knowledge through an engaging gamified experience.

- MCQ-based testing with real-time feedback.
- Competitive timers to simulate exam conditions.
- Seamless synchronization via Firebase Cloud.

Problem Statement

"Traditional assessment methods are often static and provide delayed feedback."

Students face difficulties in:

- Finding interactive platforms for quick self-evaluation.
- Managing time effectively during competitive tests.
- Tracking their progress and identifying weak areas.
- Accessing learning tools across different devices.

Project Objectives

Development

Create a high-performance web app using Flutter for a unified experience.

Engagement

Integrate timer-based logic to enhance students' competitive performance.

Analytics

Implement real-time score tracking and historical performance data.

Smart Features

Live Result Calculation
Instantaneous score updates after each question submission.

Secure Authentication
Firebase-powered user signup and login protection.

Categorized Content
Quizzes organized by subjects and difficulty levels.

Countdown Timers
Time limits per question to prevent cheating and test speed.

Responsive UI
Optimized layouts for desktop, tablet, and mobile screens.

Cloud History
All quiz history is stored securely in the cloud.

Technical Specifications

Component	Technology Stack
Framework	Flutter Web / Dart
Frontend	HTML5 / CSS3 / JavaScript
Database	Firebase Firestore
Auth	Firebase Authentication
Hosting	Firebase Hosting

Flutter Firebase Tech Stack

Methodology: Prototyping Model

We use an iterative development cycle to refine the user experience based on constant feedback.

Gathering	System Design	Prototyping	Integration	Validation
Week 1-2	Week 3	Week 4-5	Week 6-7	Week 8

Project Scope

In-Scope

- Development of Admin Dashboard.
- User Scoreboards & Leaderboards.
- Real-time database sync via WebSocket.
- Support for multiple quiz categories.

Out-of-Scope

- Native iOS/Android App (Current focus: Web).
- Video-based proctoring/surveillance.
- Offline mode functionality.

Expected Outcomes

85%

Increase in Student engagement and self-study frequency.

0

Latency in score calculation using Firebase Cloud triggers.

A scalable and robust platform that supports the Faculty of Computing's vision for a digital-first campus.

Project Roadmap

Phase 1: Requirements & Design (Week 1-3)

Phase 2: Firebase & Backend Setup (Week 4)

Phase 3: Core Dev & UI Polish (Week 5-7)

Phase 4: UAT & Final Submission (Week 8-9)

Final Conclusion

By leveraging **Flutter and Firebase**, this project provides a smart, efficient, and user-friendly solution to academic assessment. It empowers students at IUB with tools that make learning interactive and measurable, aligning with global EdTech standards.

Thank You

Questions & Feedback are welcome!

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Image Sources

https://crystalpng.com/wp-content/uploads/2024/10/IUB-logo.png

Source: crystalpng.com

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Source: codecanyon.net

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Source: potomac.edu

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